MILITARY SPECIFICATION

TERMINAL, TELEGRAPH, TRANSISTORIZED VOICE FREQUENCY CARRIER

1. SCOPE

1.1 This specification covers objectives and technical requirements for multichannel frequency shift telegraph terminals capable of operating within normal radio and wire communication channels employed in Naval Communications Systems and Facilities.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

- MIL-Q-9858 - Quality Program Requirements.
- MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
- MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
- MIL-H-46855 - Human Engineering Requirements for Military Systems, Equipment and facilities.

STANDARDS

MILITARY

- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attrubutes.
- MIL-STD-188 - Military Communication System technical Standards.
- MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
- MIL-STD-470 - Maintainability Program Requirements for Systems and Equipments.
- MIL-STD-471 - Maintainability Demonstration.
- MIL-STD-785 - Requirements for Reliability Program.
- MIL-STD-781 - Reliability Tests Exponential Distribution.
- MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

HANDBOOKS

MILITARY

- MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment.
- MIL-HDBK-472 - Maintainability Prediction.

PUBLICATIONS

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NAVSHIPS 94324 - Maintainability Design Criteria Handbook.
NAVSHIPS 94501 - Reliability Design Criteria Handbook.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEFENSE COMMUNICATIONS AGENCY

Circular 330-175-1

(Application for copies should be addressed to the Defense Communications Agency, General Offices Building 12, NSC, 8th. and S. Courthouse Road, Arlington, Va. 22204.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning production, a sample shall be tested as specified in 4.3.

3.2 Design objectives. The equipment shall be such as to attain the basic design objectives of MIL-E-16400 except that the Bureau of Naval Weapons reliability requirements and the magnetic characteristics paragraphs are not applicable.

3.2.2 Reliability.

3.2.2.1 Quantitative reliability requirements. The Voice Frequency Carrier Telegraph (VFCT) terminal shall have a specified mean-time-between failure (MTBF) (Oo) of 10,000 hours per channel. The MTBF for any piece part, modules, or subassembly whose failure would result in the failure of two or more telegraph channels shall be given by the formula MTBF $\geq$ 10,000 hours x CF. CF is the number of channel that will fail if the piece-part, module, or subassembly fails.

3.2.2.2 Reliability program plan. The contractor shall prepare for approval a reliability program plan in accordance with the requirements of this specification and the requirements of MIL-STD-785. The reliability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the reliability program. The reliability program plan shall address the applicability or nonapplicability of each paragraph of MIL-STD-785. A preliminary reliability program plan shall be submitted as part of the technical proposal. A detailed reliability program plan shall be submitted to the procuring activity thirty days after contract award, for Government approval, with an information copy to NAVELEX Code 047.

3.2.2.3 Reliability prediction. A design reliability prediction based on stress factors and part population shall be performed in accordance with MIL-HDBK-217. The prediction shall be based on the operational requirements of this specification. The reliability mathematical model and reliability block diagram shall be submitted with the reliability prediction report. Mechanical parts shall be included in the prediction. The source and applicability of all failure data used for the prediction shall be documented and applicability of all failure data used for the prediction shall be documented and justified prior to use. The results of the prediction shall be compared to the specified MTBF and where the predicted values do not achieve the requirements, the contractor shall accomplish such

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changes in design as are necessary to increase the predicted values sufficiently to acheive the requirements. An initial report shall be submitted at least ninety days prior to preproduction testing to the procuring activity, with an information copy to NAVELEX Code 047 on the procedures, results and conclusions of the reliability prediction.

3.2.2.4 Reliability design guides. The contractor shall make maximum use of the design guides contained in NAVSHIPS 94501, Reliability Design Handbook.

3.2.2.5 Reliability monthly letter reports. The contractor shall submit a monthly letter report describing the progress and results of all reliability activity performed during the reporting period and the reliability activities planned for the next reporting period. Reports on the procedures, results and conclusions for the reliability activities of 3.2.2 and the reliability program plan shall be submitted with the reliability monthly letter report. Reports shall be submitted within ten days after the end of each reporting period to the procuring activity with a copy to NAVELEX Code 047.

3.2.2.6 Reliability demonstration tests. The reliability testing shall be in accordance with 4.6.

3.2.3 Maintainability.

3.2.3.1 Quantitative maintainability requirement. The VFCT terminal shall have an equipment repair time (ERT) of no greater than 25 minutes.

3.2.3.2 Maintainability program plan. The contractor shall prepare for approval a maintainability program plan in accordance with the requirements of this procurement and the requirements of MIL-STD-470. The maintainability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the maintainability program. The maintainability program plan shall address the applicability or non-applicability of each paragraph of MIL-STD-470. A preliminary maintainability program plan shall be submitted as part of the technical proposal. A detailed maintainability program plan shall be sumitted to the procuring activity, thirty days after contract award, for Government approval, with an information copy to NAVELEX Code 047.

3.2.3.3 Maintainability prediction. A maintainability prediction shall be performed using procedure II of MIL-HDBK-472. The contractor shall document and justify all assumptions and the applicability of all data used in the prediction. The prediction shall also include mechanical parts. The results of the prediction shall be compared to the required ERT and where the predicted values do not achieve the requirements the contractor shall accomplish such changes in design as are necessary to improve the predicted values sufficiently to achieve the requirements. A report shall be submitted ninety days prior to preproduction testing to the procuring activity, with an information copy to NAVELEX Code 047 on the procedures, results, and conclusions of the maintainability prediction.

3.2.3.4 Maintainability design guides. The contractor shall make maximum use of the design guides contained in NAVSHIPS 94324, Maintainability Design Criteria Handbook.

3.2.3.5 Maintainability monthly letter report. The contractor shall submit a monthly letter report describing the progress and results of all maintainability activity performed during the reporting period and the maintainability activities planned for the next reporting period. Reports on the procedures, results and conclusions for the maintainability activities of 3.2.3 and the maintainability program plan shall be submitted within ten days after the end of each reporting period to the procuring activity with a copy to NAVELEX Code 047.

3.2.3.6 Maintainability demonstration test. The maintainability requirements of the equipment shall be demonstrated in accordance with 4.7.

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3.2.4 Human factor engineering. In order to realize the man-machine relationship from the stand-point of both operation and maintenance, the contractor shall comply with the human factor design standards of MIL-STD-1472 and the requirements specified herein.

3.2.4.1 Human engineering plan. A Human engineering plan, the program plan and drawing approval in the appendix shall be submitted in accordance with MIL-H-46855 as part of the contractor's original bid and revised and resubmitted sixty days after contract award to the procuring activity with an information copy to NAVELEX Code 047.

3.2.4.2 Guidance meeting. Within thirty days after contract award, the contractor shall arrange a meeting with appropriate representatives of the procuring activity, including NAVELEX Code 047, for the purpose of obtaining guidance on complying with the human factor requirements of this specification. The meeting shall be directed toward achieving mutual understanding of information contained in the contractor's proposal and the requirements imposed by this specification.

3.2.4.3 Equipment requiring human engineering review. A list of all equipment requiring human engineering design detail for operation and maintenance shall be prepared as required by MIL-H-46855 and submitted to the procuring activity with an information copy to NAVELEX Code 047. This list, to be initially submitted six months after contract award and updated and resubmitted eighteen months after contract award shall separate the equipments into operator and maintainer man-machine interfaces and it shall describe, as a minimum, which front panel configurations and equipment interior views should be reviewed for conformance to the human engineering design criteria of this specification.

3.2.4.4 Auditory alarms. The requirements of MIL-STD-1472 for auditory displays shall be observed in the design of a summary audible alarm system. This system shall respond to blown fuzes as required by 3.17.4.1.3 and to other alarms indicated by alarm lamps on the front panels. The audio warning device shall have an indicating lamp which is not extinguished by the disabling switch, and it shall have a disabling switch with automatic reset circuitry.

3.2.4.5 Human engineering design verification. A human engineering design verification plan to test the equipment and system of paragraph 1.1 shall be prepared in accordance with the requirements of design verification-test and evaluation, and the failure analyses of MIL-H-46855, 4.3 of this specification, and the Human Factor and Maintenance Checklist to be supplied by NAVELEX. This plan shall be submitted to the procuring activity with an information copy to NAVELEX Code 047 for review and approval sixty days prior to the test date.

3.2.4.6 Human engineering progress reports. Progress reporting on human engineering work shall conform to the requirements listed in the appendix of MIL-H-46855 except that the progress reports shall be submitted at least twice to the procuring activity with an information copy to NAVELEX Code 047. The first report is to be submitted not less than ninety days prior to the human engineering design verification testing of paragraph 3.2.4.5 and the second report not later than thirty days after testing.

3.2.4.7 Change requests. In the event the contractor wishes to make a change in the approved human engineering plan, 3.2.4.1, the approved panel drawings, 3.14, or the approved human engineering verification plan, 3.2.4.5 a change request shall be submitted to the procuring activity with an information copy to NAVELEX Code 047 for approval. This request, to be submitted at least thirty days prior to the effective date of the change, shall state what the change is, why it is required, and its effect on the system, equipment or facilities.

3.3 Definitions. The definitions of terms used in this specification shall be in accordance with MIL-E-16400.

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3.4 Modular construction for equipment. Modular construction shall be in accordance with MIL-E-16400.

3.5 Parts. Parts shall be in accordance with MIL-E-16400.

3.6 Material. Material shall be in accordance with MIL-E-16400 except that radioactive material shall not be used in any form and the following phrase and paragraphs do not apply:

(a) The magnetic permeability phrase under metals.
(b) Equipment subject to sea water immersion.
(c) Magnetic materials.

3.7 Coatings and treatment. Coatings and treatment shall be in accordance with MIL-E-16400.

3.8 Safety. In order to realize safe operation and maintenance of the equipment of paragraph 1.1 from the standpoint of safety for personnel and equipment, the contractor shall comply with the safety design standards in MIL-E-16400. In addition, the following requirements shall be met.

3.8.1 Voltage below 30 volts (rms or dc) shall be used whenever possible.

3.8.2 All access doors which may be opened to expose voltages in excess of 30 volts during operation and maintenance shall be clearly marked "DANGER - VOLTAGE DANGEROUS TO LIFE." The letters on the marking shall be Gothic capitals, clearly legible, color white with red background (American Standard Z 35.1 - 1959).

3.8.3 The voltage terminals for all voltages in excess of 30 volts which are not deactivated by an interlock circuit when an access door is opened shall be protected by barriers which are clearly marked "CAUTION - HIGH VOLTS." The letters on the marking shall be Gothic capitals, clearly legible, color black with yellow background.

3.8.4 The interlock over-ride switch shall be located on the front of the equipment. It shall be possible when the switch is in the over-ride position to open the rear access door without interrupting the power to the unit. In the non-over-ride position whenever the rear door is open the power to the unit shall be interrupted.

3.9 Environmental conditions. The equipment shall meet the requirements of MIL-E-16400 for temperature (Class 4 of table II), and for humidity.

3.10 Electrical design. Electrical design shall be in accordance with MIL-E-16400 except that the preferred circuits and the internal supply voltages paragraphs do not apply.

3.10.1 Blown fuse indicators and an alarm shall be required.

3.10.2 The equipment shall operate with primary power line voltages of 115 volts or 230 volts.

3.10.3 The equipment shall operate with primary power line frequencies of 48 to 400 cycles.

3.11 Maintenance Design. Maintenance design shall be in accordance with MIL-E-16400.

3.11.1 One double receptacle with ground terminals shall be provided on the front and rear of the cabinet located near the bottom and shall be entirely separate from ac power to the VFCT terminal.

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3.12 Mechanical Design. Mechanical design shall be in accordance with MIL-E-16400 except that the following paragraphs do not apply.

a. Shock
b. Vibration
c. Loading
d. Enclosures
e. Size
f. Handling

3.13 Thermal design. Thermal design shall be in accordance with MIL-E-16400 except that the heat removal method shall be as specified for equipment which is not intended for shipboard use.

3.14 Controls, indicators and panel layout. Controls, indicators and panel layout shall be in accordance with MIL-E-16400, except that time meters shall not be required.

3.15 Supplementary requirements. Supplementary requirements shall be in accordance with MIL-E-16400 except that the Bureau of Naval Weapons contracts paragraph does not apply.

3.16 Electromagnetic interference emission, susceptibility, and systems self compatibility. The equipment shall conform to class IC of MIL-STD-461.

3.17 Functional requirements.

3.17.1 Telegraph terminal. The telegraph terminal shall be completely solid state. All active or rectifying electrical elements shall be semiconductor devices. Electromechanical relays shall not be used. Vacuum tubes shall not be permitted. The terminal shall be capable of operating over a communications channel with the AN/FGC-60 on AN/FCC-66-72 or AN/FCC-89. The input and output devices shall conform to 3.17.3.1 and 3.17.3.2.

3.17.1.1 All terminals shall be completely wired, harnessed and factory adjusted so as to be completely operational in the configuration specified after initial on-site installation and alignment.

3.17.2 Each telegraph terminal shall be based on modular building block techniques. It shall be capable of being configured in any of the arrangements described herein or in any combination or portion thereof.

3.17.2.1 Sixteen narrow band telegraph channels within a bandwidth of 375 to 3025 Hz.

3.17.2.1.1 Configurations.

(a) 16 channel full duplex non-diversity
    (1) 16 channel non-diversity, send only
    (2) 16 channel non-diversity, receive only
(b) 16 channel full duplex dual and quadruple diversity

3.17.2.2 Non-diversity, dual diversity, or quadruple diversity.

3.17.2.2.1 Dual diversity as used in this specification shall depend on the nature of the transmission media. Dual diversity shall be accomplished either by space diversity or frequency diversity operation of the transmission media.

3.17.2.2.2 Frequency diversity as applied to the device for this specification is defined as the transmission of identical information on two different channels within a single 3KC band. This function is defined as in-band diversity.

3.17.2.2.3 Quadruple diversity is the combination of diversity operation on the transmission media and in-band diversity.

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3.17.2.3 All routine test and adjustment points shall be accessible from the front of the terminal without removal of the modules from the equipment shelves. One extender module shall be furnished with each terminal for maintenance purposes. When not in use the extender module shall be housed in an allocated space (accessible from the front) in the terminal in the same manner as the other modules. When in use the extender module shall provide the electrical connections and mechanical support necessary to permit moving the module forward an amount at least equal to its depth and operate the module as part of the VFCT terminal.

3.17.3 The telegraph terminal shall receive from and transmit to telegraph loops as follows:

3.17.3.1 The transmit section of each telegraph terminal shall accept input telegraph signals in any of the following modes.

(a) 60 milliampere, 130 volt neutral (high level neutral).
(b) 20 milliampere, 60 volt polar (high level polar).
(c) 1 milliampere, 6 volt polar (low level polar).

3.17.3.1.1 The input characteristics shall be as specified in the digital transmission reference systems of MIL-STD-188 except that the terminal shall be required to accept input signals containing up to 40 percent total peak telegraphic distortion. It is not required that output signals from the transmit section contain less distortion than the corresponding input signals.

3.17.3.1.2 The mode shall be independent for each telegraph loop and selectable by strapping appropriate terminals which are readily accessible.

3.17.3.1.3 The safety factor for high level modes shall be at least plus or minus 2.5; that is when a channel is strapped for x ma and y volts no damage shall result if the input signal is plus or minus 2.5 x ma and plus or minus 2.5 y volts.

3.17.3.1.4 The safety factor for low level polar shall be plus or minus 6 for both current and voltage.

3.17.3.2 The receive section of each telegraph terminal shall generate output telegraph signals in any of the following modes:

(a) 60 milliampere, 130 volt neutral (high level neutral)
(b) 20 milliampere, 60 volt polar (high level polar)
(c) 1 milliampere, 6 volt polar (low level polar)

3.17.3.2.1 The output characteristics shall be as specified in the digital transmission reference system of MIL-STD-188 for high performance facilities except that the rise and fall times of the signal shall be within 1 percent of each other and between 5 percent and 10 percent of the unit interval. Suitable current and voltage adjustments and protection shall be provided for each output telegraph loop.

3.17.3.2.2 The mode shall be independent for each loop and selectable on a strapping basis.

3.17.4 The telegraph terminal shall receive from and transmit to communications channels whose transmission media may be metallic circuits, microwave or hf radio.

3.17.4.1 When a terminal is in its normal operating environment and performing within the requirements of this specification, the performance of the terminal shall not be degraded beyond the limits of this specification if any normal operational adjustments are made to the terminal within the requirements of this specification, such as signal level, polarity or modulation rate.

3.17.4.1.1 It shall be possible to interchange keyers and convertors so as to change a terminal

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from full duplex to simplex or vice-versa (32 send or 32 receive) between non-diversity terminals after the terminals have been installed and operating on site. No equipment or component wiring changes shall be required. For example, if two 16-channel non-diversity duplex terminals are installed and operating it shall be possible to remove the keyers from one of the cabinets and replace the keyers with the corresponding convertors from the other cabinet without making any wiring changes or additions to either terminal equipment other than retermination of signal lines. The terminals in the changed configurations shall meet all applicable requirements of this specification without having to make any adjustments or alignments. No damage shall occur if a keyer is plugged into a slot that is externally wired for a convertor or vice-versa.

3.17.4.1.2 It shall be possible to operate a 16-channel diversity or non-diversity terminal on a sub-system basis of 4 keyers and 4 convertors for non-diversity terminals and 4 keyers, 8 convertors, 4 combiners and twinning units for diversity terminals per subsystem in any of the following configurations:

(a) 4 separate subsystems of 4 channels each
(b) 2 separate subsystems of 4 channels each and one separate subsystem of 8 channels
(c) 2 separate subsystems of 8 channels each
(d) 1 separate subsystem of 12 channels and 1 separate subsystem of 4 channels
(e) 1 separate system of 16 channels

3.17.4.1.2.1 The configurations in 3.17.4.1.2 shall be accomplished by setting switches on the front of the terminal. In any of these configurations each subsystem of the terminal shall operate within the requirements of this specification without requiring any adjustments or alignment. There shall be independent switches for the transmit section and receive section.

3.17.4.1.2.2 Each configuration in 3.17.4.1.2 shall interface with a landline or microwave link. The interface provided in any mode of operation shall be: 600 ohm balanced with center tap available; minimum return loss of 26 db over the frequency range 370 to 3040 hz; 40 db longitudinal current suppression.

3.17.4.1.2.3 The single line drawings, Figures 1 and 2, depict the output circuitry of the keyers and the input circuitry of the converters of each 16-channel diversity and non-diversity telegraph terminal and may be used to interpret the requirements in 3.17.4.1.2.

3.17.4.1.2.4 All terminals shall be wired and harnessed and otherwise furnished so as to be operable under the subsystem concept specified in 3.17.4.1.1 except that the number of amplifiers to be furnished with each terminal shall be as follows unless otherwise specified:

(a)	Non-diversity configurations	
	(1)	16 channel sent (2 spares)
	(2)	16 channel receive (1 on line, 1 spare)
	(3)	16 channel duplex (1 on line, 1 spare)
(b)	Diversity	
	(1)	16 channel duplex (2 on line, 1 spare)

3.17.4.1.2.4.1 It shall be possible on a strapping basis to insert the spare line amplifier in the composite tone unput and output lines of the terminal.

3.17.4.1.3 Blown fuse indicators shall be required; each module shall be internally fused. In the event of a failure of a power supply that is common to more than 2 modules, a fuse indicator shall be activated. Each fuse indicator shall be capable of activating a common audible alarm when a failure occurs. All fuse indicators and fuses shall be located on the front panel.

3.17.4.2 When the composite output signal of the transmit section, in any configuration is connected on a back to back basis to: (1) the first diversity receive section only; (2) the second diversity receive section only; (3) the first and second diversity receive sections, the following requirements shall be met

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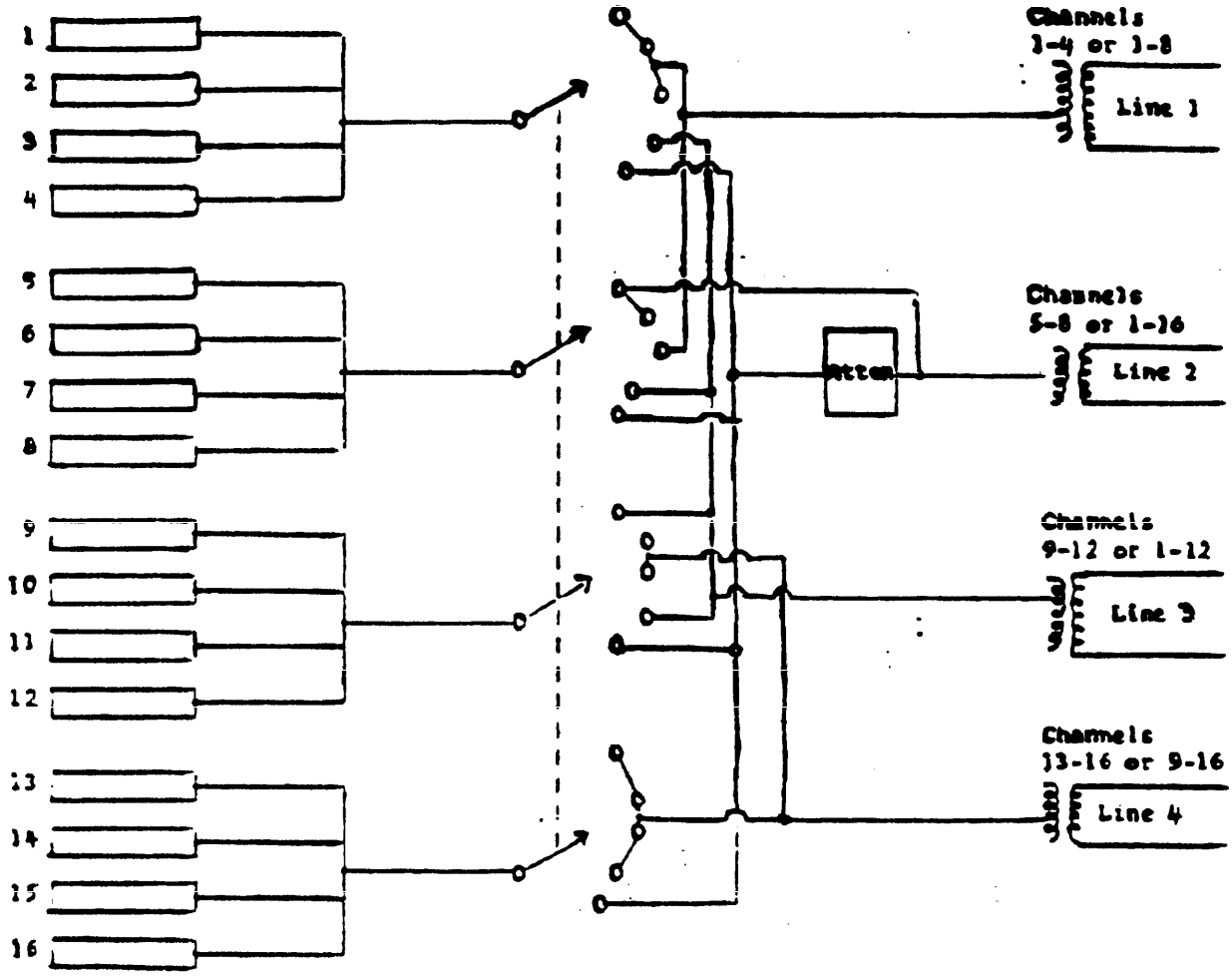


Figure 1 - Output Circuitry of Keyers

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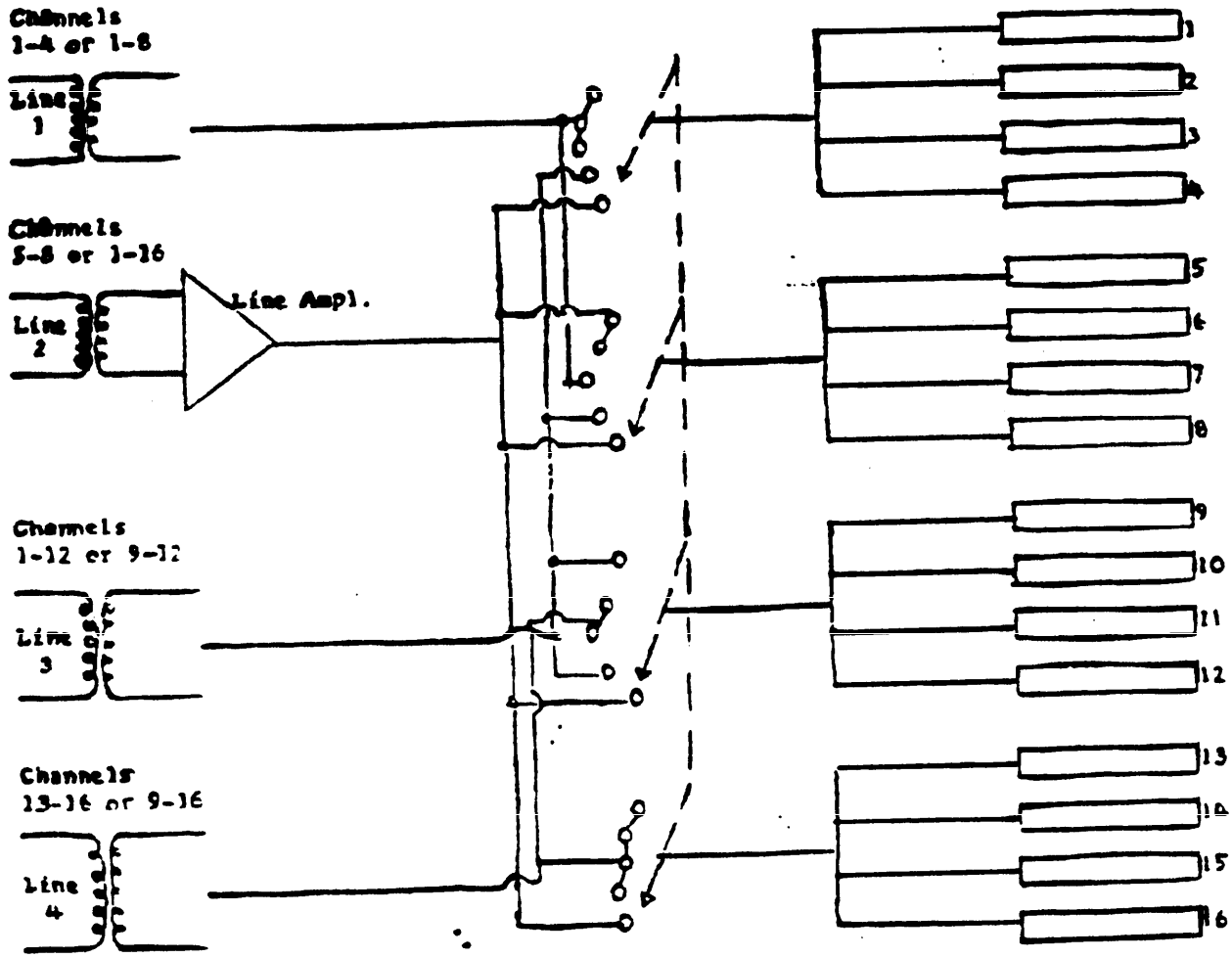


Figure 2 - Input Circuitry of Converters.

Note: For diversity configurations a second receiving circuitry identical to above shall be provided.

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3.17.4.2.1 When all output channels are continuously keyed with 90 baud random signals (the input to any channel is completely independent of the input to any other channel) with less than 1 percent distortion in any of the modes specified in 3.17.3, the telegraph signals appearing at the output of each receive channel, shall have less than 5 percent total peak telegraphic distortion when all keyers are adjusted so that the input signals to each converter are: (1) All at the level of -40 dbm; (2) All at the level of -10 dbm; (3) All at any level in the range of -40 dbm to -10 dbm. Under the same conditions the total peak telegraphic distortion shall not exceed 10 percent when input signals to the converters are in the range of -50 dbm to -40 dbm and -10 dbm to 0 dbm. When any channel is idled out, the adjacent channel interference measured on the loop side of the idle channel filters shall not exceed -60 dbm.

3.17.4.2.1.1 When the terminal is operating with the input signals to each converter at -20.7 dbm, if any one of the transmit channels is attenuated -30db, the output telegraph signal from the corresponding receive channel shall not have more than 40 percent total peak telegraphic distortion.

3.17.4.2.1.2 When the transmit section is connected to the first and second diversity receive sections, either receive section shall be capable of accepting a -50 db fade of the composite signal without increasing the total peak telegraphic distortion beyond the specified limits.

3.17.4.2.2 When all input channels are keyed with the same 90 baud random signals in any of the modes specified in 3.17.3 the differential delay between any two output channels from the convertors shall not exceed 500 microseconds when all channel delay adjustments are set to zero delay.

3.17.5 All the necessary parts that comprise the following shall be housed in one standard 19 inch cabinet (similar to a CY-597) with a maximum over-all height of the cabinet of 80 inches.

(a) 16 channel, full duplex, ND.
  (1) 16 channel, ND, Send only.
  (2) 16 channel, ND, Receive only.
(b) 16 channel, full duplex, diversity.

3.17.5.1 The UFCT terminal shall be completely installable and maintainable from the front of the cabinet.

3.17.5.2 Rear access door shall be provided on all cabinets.

3.17.5.3 All configurations of telegraph terminals shall be comprised of the following items. The number of items shall be in accordance with the configuration specified. The physical configurations of the parts or modules need not coincide precisely with the description, provided that all other requirements are satisfied.

3.17.5.3.1 Keyers. All keyers shall be narrow band (as follows) unless otherwise specified. One frequency shift keyer per transmit channel shall be required for all configurations. A keyer shall accept an input signal at any modulation rate up to 90 baud at any of the modes specified in 3.17.3.1 and convert it to an audio signal as specified herein for nondiversity configurations or as specified in paragraph 3.17.5.3.4 for diversity configurations.

Channel Number	Mark Frequency (Hz)	Center Frequency (Hz)	Space Frequency (Hz)	Sending Signal Sense (Non-diversity)
1	382.5	425	476.5	-
2	552.5	595	637.5	-
3	722.5	765	807.5	-
4	892.5	935	977.5	-
5	1062.5	1105	1147.5	-

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Channel Number	Mark Frequency (Hz)	Center Frequency (Hz)	Space Frequency (Hz)	Sending Signal Sense (Non-diversity)
6	1232.5	1275	1317.5	-
7	1402.5	1445	1487.5	-
8	1572.5	1615	1657.5	-
9	1742.5	1785	1827.5	-
10	1912.5	1955	1997.5	-
11	2082.5	2125	2167.5	-
12	2252.5	2295	2337.5	-
13	2422.5	2465	2507.5	-
14	2592.5	2635	2677.5	-
15	2762.5	2805	2847.5	-
16	2832.5	2975	3017.5	-

3.17.5.3.1.1 Adjacent channel separation shall be 170 Hz.

3.17.5.3.1.2 The frequency deviation from the center frequency shall be plus or minus 42.5 Hz.

3.17.5.3.1.3 Provision shall be made for inverting the output signal Mark-Space frequency assignment with respect to the input signal for any keyer as specified in 3.17.5.3.1 and 3.17.5.3.4. This shall be accomplished by setting a switch on the keyer that is readily accessible but not located on the front of the module.

3.17.5.3.1.4 The output level shall be continuously adjustable over a minimum range of 0 dbm to -30 dbm with a single control.

3.17.5.3.1.5 The mark and space output signal levels for any one keyer shall be within 1.5 dbm of each other.

3.17.5.3.1.6 The harmonic content output of any keyer shall be 60 db below the output signal.

3.17.5.3.1.7 All keyers shall have the same physical dimensions and mounting requirements. The units shall be identical for any operating frequency and frequency-shift except for the modular plug-in subassemblies which determine the operating frequency and shift. Each keyer shall provide for mounting a channel designator nameplate on the front. The nameplate shall be furnished with the frequency and frequency-shift elements and shall designate the mark, space and center frequencies.

3.17.5.3.1.8 Readily accessible strapping terminals shall be provided on each keyer in order to select the mode of input signal as specified in 3.17.3.1.

3.17.5.3.1.9 Test monitor jacks for all essential test points shall be provided on the front panel of each keyer. The minimum acceptable front panel monitor jacks shall be as follows:

(a) Input telegraph signal
(b) Keyer audio output signal
(c) Input signal to bandpass filter

3.17.5.3.1.10 Each keyer shall be capable of transmitting continuous mark, continuous space, of accepting keying from the external signal line or to disable the keyer output without removal of power from the unit. This shall be accomplished by setting a switch located on the front panel of the module.

3.17.5.3.1.11 No adjustments shall be required except for setting the output level frequency and keyover point. The frequency shall be adjustable to within plus or minus 0.1 Hz from the specified frequency. The keyover point shall be adjustable plus or minus 5 percent of the nominal

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mark level from the specified keyover point. The frequency and sensitivity adjustments shall be readily accessible but not located on the front panel of the keyer. The output level adjustment shall be located on the front panel of the keyer.

3.17.5.3.1.12 The input shall be electrically floating so that either or neither input terminal may be grounded at the option of the user by strapping appropriate terminals which are readily accessible.

3.17.5.3.1.13 The frequency characteristics of any channel at the mark and space frequencies shall be as follows.

3.17.5.3.1.13.1 The frequency tolerance shall be plus or minus 0.025 percent at all temperatures and voltages within the specified operating ranges except that the maximum deviation shall not exceed plus or minus 0.5 Hz.

3.17.5.3.1.13.2 The short term (eight hours) and long term (six months) frequency stability shall be within the limits of the frequency tolerance in 3.17.5.3.1.13.1.

3.17.5.3.1.14 The jitter shall be less than 2 percent for any frequency.

3.17.5.3.1.15 The keyover point shall be 30 ± 1 ma for high level neutral and 0 ± 0.167 ma for high level polar. The low level polar keyover point shall be as described in receiver sensitivity of MIL-STD-188. That is, when the mark and space sense of the input signal changes, the output mark and space frequency shall change accordingly at the keyover point. The output mark and space frequencies shall be mutually exclusive of each other.

3.17.5.3.2 Convertor/Loop Keyer. All convertors shall be narrow band. Two frequency shift convertors and one loop keyer per receive channel shall be required for all diversity configurations. The frequency assignment shall be identical to that of the keyers as specified in paragraphs 3.17.5.3.1 and 3.17.5.3.4. Each convertor/loop keyer shall accept an input of 16 tone composite audio signal, select the assigned channel frequency and discriminate against the unwanted channels. The selected channel frequency is converted to subparagraphs. The power requirements (loop battery) for all specified modes of telegraph signals shall be provided wholly by each loop keyer and AC shall be directly from the primary AC power line. The ripple voltage shall not exceed 1 percent peak to peak. Provision shall be made for use of an optional external battery for telegraph loops. Each loop keyer battery shall be capable of providing 60 milliamperes into a varying output load of 50- to 1800 over the primary alternating current voltage and frequency variation specified in MIL-E-16400.

3.17.5.3.2.1 Provision shall be made for inverting the mark and space sense of the output signal with respect to the input signal for any convertor. This shall be accomplished by setting a switch on the convertor that is located on the front of the module.

3.17.5.3.2.2 The convertor/loop keyer shall perform within the requirements of this specification without readjustment when receiving a signal from the corresponding transmitter at any level in the range from -50 dbm to 0 dbm or when the signal is varied within this range at a rate encountered in fast fading hf radio.

3.17.5.3.2.3 The bias distortion shall be continuously adjustable over a minimum range of ± 10 percent at any rate between 45 and 90 baud. When the input signal is anywhere within its rated limits and contains less than 1 percent bias distortion the output from the convertor shall be adjustable so as to produce less than 1 percent bias distortion.

3.17.5.3.2.3.1 When the frequency of the input signal (at any rate up to 90 baud) varies ± 5 cycles from its assigned values, the total peak telegraphic distortion from the convertor/loop keyer shall not exceed 15 percent.

3.17.5.3.2.4 The time delay of the output telegraph signal shall be continuously adjustable over a minimum range of zero to 7 milliseconds.

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3.17.5.3.2.5 All convertors and loop keyers shall have the same physical dimensions and mounting requirements. The units shall be identical for any operating frequency and frequency shift except for the modular plug-in subassemblies which determine the operating frequency and shift. Each convertor shall provide for mounting a channel designator nameplate on the front. The nameplate shall be furnished with the frequency and frequency shift elements and shall designate the mark, space and center frequencies.

3.17.5.3.2.6 Readily accessible strapping terminals shall be provided on each loop keyer in order to select the mode of output signal as specified in 3.17.3.2.

3.17.5.3.2.7 Test monitor jacks for all essential test points shall be provided on the front panel of each convertor/loop keyer. The minimum acceptable front panel jacks shall be as follows:

(a) Audio signal on loop side of telegraph channel bandpass filter
(b) Output signal from discriminator
(c) Output telegraph signal

3.17.5.3.2.8 The output shall be electrically floating so that either or neither output terminal may be grounded at the option of the user by strapping appropriate terminals which are readily accessible.

3.17.5.3.3 Diversity combiners. One combiner per pair of receive convertors shall be required for the diversity configuration. Convertor twinning for frequency diversity shall be as shown in 3.17.5.3.4. Provisions shall be made for selecting the mode of operation by a switch mounted on the front panel of the combiner module marked as follows (a) dual diversity, and (b) quadruple diversity. It shall not be necessary to make any adjustments when the mode is changed.

3.17.5.3.3.1 The combining operation shall be correlated with the associated convertors in a manner so as to provide to the maximum extent feasible an output signal based on the best momentary signal to noise ration for the information bit being detected. Under deteriorating signal to noise conditions, the combiner shall provide a diversity circuit improvement of from 2 to 3 db over nondiversity. This improvement shall be demonstrated in the following manner:

(a) With a constant noise level and unequal varying signals in the channels to be combined.
(b) With signals constant and varying unequal noise in the channels to be combined.

3.17.5.3.3.2 All diversity combiner modules shall be physically and electrically identical. Each combiner shall perform within the requirements of this specification when the input signals to the convertor are in the range of 0 to 50 dbm.

3.17.5.3.3.3 Test monitor jacks for all essential test points shall be provided on the front panel of each diversity combiner. The minimum acceptable front panel monitor jacks shall be as follows:

(a) First and second diversity input signals
(b) Combined output signal

3.17.5.3.4 Keyer twinning panels for local and remote shall be furnished for each diversity terminal. Keyers shall be twinned on the adjacent numerical channels shown herein for example, channels 1 and 2, 3 and 4. When twinning is employed the even numbered input channel shall be keyed simultaneously with its associated odd numbered channel, and even number channel output shall be combined with the associated odd channel in comparator circuits to produce a single output from the odd channel. The ac telegraph keying loops on even channel inputs and output shall be open circuited.

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Channel Number	Center Frequency Diversity	Sending Signal Sense Diversity
1	1785	-
2	425	-
3	1955	-
4	595	-
5	2125	-
6	765	-
7	2295	-
8	935	-
9	2465	-
10	1105	-
11	2635	-
12	1275	-
13	2805	-
14	1445	-
15	2975	-
16	1615	-

3 17.5.3.5 Amplifier

3 17.5.3.5.1 Frequency response shall be flat plus or minus 2 db from 300 Hz to 6000 Hz.

3 17.5.3.5.2 The input and output impedance shall be 600 ohms balanced, with a minimum return loss of 26 db over the frequency range 300 to 6000 Hz The longitudinal current suppression shall be at least 40 db below the input or output signal. Transformer coupling shall be required.

3.17.5.3.5.3 The gain shall be continuously adjustable, from 0 db to 35 db when the input level is between - 50 dbm and 0 dbm. The output level shall not exceed + 14 dbm ± 1db.

3.17.5.3.5.4 When the input signal is varied over the range 300 Hz to 6000 Hz at any level between -50 dbm and 0 dbm the maximum value of any harmonic generated by the amplifier shall be at 40 db below the output level for any level up to + 10 dbm. The hum and noise level shall be at least 50 db down from 0 dbm output.

3.17.5.3.5.5 Test monitor jacks for all essential test points shall be provided on the front panel of each amplifier. The minimum acceptable front panel jacks shall be as follows:

(a) Input signal
(b) Output signal

3.17.5.3.6 Attenuator One attenuator per transmit terminal shall be required. The output from the keyers shall terminate at a common nodal entry point to the variable attenuator.

3.17.5.3.6.1 The composite output level of the attenuator shall be adjustable over a minimum range of 0 db to-30 db. Continuously variable attenuators shall not be permitted.

3.17.5.3.6.2 The output impedance of the attenuator shall be 600 ohms balanced with the center tap available.

3.17.5.4 Test panel. A test panel shall be provided for each telegraph terminal unless otherwise specified in the contract or order. The test panel shall be mounted in the cabinet in a manner that will permit viewing of the instruments from the front of the cabinet. The panel and instruments shall be completely wired to all necessary points within each terminal so that the operator may perform all the desired tests by setting appropriate switches The panel and switching arrangement shall be designed so that simplicity and ease of operation are optimized. The purpose

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of the test panel is to provide the instrumentation necessary for test and alignment of the telegraph terminal. Each instrument shall be under switch control so that the operator can electrically place the instrument at the desired points within the terminal. Each instrument shall have two test jacks associated with it so that the instrument may be used for testing at points other than those specified below and one additional test jack for oscilliscope monitoring. Dummy loads shall be supplied where appropriate. The following instruments and corresponding switch selectable test points shall be required.

3.17.5.4.1 Instrument - VU meter. A standard volume level indicating meter (two required for diversity configurations) shall be provided with a step type attenuator to measure input levels from -40 to + 10 dbm. The attenuator shall be in increments of 2 db steps. The input shall also be provided with a balanced bridging of 7500 ohms minimum resistive. The meter ballistic characteristics shall be that of a standard VU meter. Diversity configurations shall employ one VU meter in each of the diversity circuits. Provision shall be provided to bridge the individual inputs of converter 3.17.5.4.1.1. Test points shall be:

(a) Each output of each keyer
(b) Composite tone out of transmit section
(c) Composite tone in and out of each amplifier
(d) Each input to each converter

3.17.5.4.2 Instrument - Volt/ammeter. A volt ammeter shall be provided and calibrated in 3 ranges to indicate the 3 modes of input and output telegraph signals, namely (-) 75-0-75, (-)25-0-25 milliamperes and -10, 0, -10 volts. A screwdriver adjustment shall be provided for zero adjustment. Protection shall be provided to prevent damage to the meter if inadvertently placed in the wrong mode.

3.17.5.4.2.1 The test point shall be each input and output telegraph loop.

3.17.5.4.3 Instrument - Delay meter. A differential delay meter calibrated from 0 to 10 milliseconds full scale deflection shall be provided to compare any two channels. A screwdriver adjustment shall be provided for zero adjustment.

3.17.5.4.3.1 Test Points shall be each pair of adjacent channels in frequency diversity mode.

3.17.5.4.4 Instrument - Dot cycle generator. The dot cycle generator shall generate reversals at 75 and 90 baud with not more than 1 percent total peak telegraphic distortion. The generator shall be switchable to the input of any or all keyers.

3.17.5.4.5 Instrument - Distortion monitor. A distortion test set shall be provided to work in conjuction with the dot cycle generator for test and alignment. The test panel shall permit the operator to place the dot cycle generator on the input to any or all keyers, connect the output of the transmit section to the input of the receive section, place the distortion test set on the ouput of any convertor and measure the output telegraphic distortion. A suitable visual indication of the distortion shall be required.

3.17.5.4.6 Back-to-Back. Provision shall be incorporated into the test panel to perfrom the following functions in any of the subsystem mode of operation.

(a) Back-to-back on an individual channel basis without interfering with the other channels.
(b) Back-to-back of all channels through all modules or components which make up a duplex system.

3.17.6 Power supply. The VFCT terminals shall be configured to operate from individual loop keyer power supplies and shall have the capability of operating from one or both power supply methods as specified in DCAC 330-175.1.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. The quality control system shall be in accordance with specification MIL-Q-9858. A current written description of the system shall be submitted to the procuring activity via the cognizant Government inspector, for system approval, sixty days prior to preproduction inspection. If preproduction inspection is not required, the written description of the quality control system shall be submitted for approval sixty days prior to production inspection. The written description shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flowcharts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the approved quality control system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the procuring activity for approval prior to use.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Tests, Test Equipment and Test Procedures. The contractor shall submit a list of proposed tests and written test procedures including test equipment and tolerance limits to be used in preproduction or first article inspection & quality conformance inspection to the procuring activity, via the Government inspector for review and approval sixty days prior to beginning tests. Each test shall be identified in accordance with 4.2 as to classification.

4.2 Classification of inspection. The method of examination and testing of the equipment fall within the following classifications:

(a) Preproduction inspection
(b) Quality conformance inspection
  (1) Production inspection
  (2) Production control inspection

4.3 Preproduction inspection. Preproduction inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification and shall include the following:

Requirement	Reference
Production inspection	4.4.1
Production control inspection	4.4.2
Temperature and humidity	3.9
Electromagnetic interference Emision, susceptibility, and system self compatability	MIL-E-16400

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4.3.1 Preproduction inspection report. Six copies of the preproduction inspection report shall be submitted within thirty days of completion of the preproduction inspection to the procuring activity via the Government inspector for review and approval. The preproduction inspection report shall indicate specification compliance limits as well as actual performance test result. The contractor shall certify that, on the basis of the report, the equipment does or does not comply with specification requirements.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects in materials. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the following:

Requirement	Reference
Surface examination	MIL-E-16400
Operating test	3, 17, 4.5.1 and MIL-E-16400

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table I and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table I - Production Control Inspection

Inspection	Inspection Paragraph	AQL%
Supply line voltage and frequency	3.10 and MIL-E-16400	10
Weights and dimensions	MIL-E-16400	10
Heat test	MIL-E-16400	10
Controls and control circuity	MIL-E-16400	10
Power	MIL-E-16400	10

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.5 Inspection procedures.

4.5.1 Operating test. The equipment shall be subjected to an operating test. The input and output signals shall be high level neutral. The operating test for the equipment shall be performed with all modules and parts in place in the cabinet of the terminal with the rear door of the cabinet closed. The operating test shall assure qualitatively the proper functioning of the equipment, including all operating controls and conformance with the safety requirements.

4.6 Reliability testing. The reliability of the VFCT terminal shall be verified and reported in accordance with MIL-STD-781. The specified MTBF shall be 10,000 hours per channel for reliability testing.

4.6.1 Demonstration phase of production reliability tests. The demonstration phase of

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reliability tests shall be conducted with test plan IV at test level A1 in MIL-STD-781. The sample size shall be four production units taken from the initial months production. The reliability requirements shall be successfully demonstrated prior to the shipment of any production equipment.

4.6.2 Production acceptance (sampling) phase of production reliability tests. The production acceptance (sampling) phase of reliability tests shall be conducted with test plan IV-A at test level (A1 in MIL-STD-781. The sample size shall be three units from each subsequent month's production. The reliability test shall be successfully completed prior to the shipment of any production lots.

4.6.3 Reliability test plan. The contractor shall prepare a detailed reliability test plan in accordance with MIL-STD-781. Test parameters and conditions shall be in accordance with the equipment operational requirements as defined in the equipment specification. Performance parameters to be measured during the reliability test shall be in accordance with the equipment specification. Debugging and burn-in prior to the reliability test shall be limited to the debugging and burn-in to be performed on all production equipment prior to delivery to the Government. No preventative maintenance shall be performed during the reliability test. At least ninety days prior to the start of reliability testing, the contractor shall submit a reliability test plan to the procuring activity for Government approval, with an information copy to NAVELEX Code 047.

4.6.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification. All failures shall be considered relevant unless determined by the procuring activity to be caused by a condition external to the equipment under test which is not a test requirement and not encountered in service. Only relevant failures shall be counted during the reliability test for an accept or reject decision.

4.6 5 Final restoration and disposition of tested equipment. Upon completion of reliability testing, the tested equipment shall be returned to production for reinspection and reconditioning as required, and may be delivered as part of the contract quantity if it meets all terms and conditions of the contract for acceptance.

4.6.6 Reliability test reports. The contractor shall submit test reports for reliability testing in accordance with MIL-STD-781. Reliability test reports shall be submitted within ten days after completion of each reliability test to the procuring activity for Government approval, with an information copy to NAVELEX Code 047.

4.7 Maintainability demonstration test. The maintainability requirements of 3.2.3.1 for the equipment shall be successfully demonstrated in accordance with method 3 of MIL-STD-471.

4.7.1 Maintainability demonstration test plan. The contractor shall prepare a detailed maintainability test plan in accordance with MIL-STD-471. Ninety days prior to the start of testing, the contractor shall submit the maintainability demonstration test to the procuring activity, for Government approval, with an information copy to NAVELEX Code 047 for Government approval prior to the start of testing.

4.7.2 Maintainability demonstration report. A final maintainability demonstration report shall be submitted in accordance with MIL-STD-471. The maintainability demonstration report shall be submitted to the procuring activity, with an information copy to NAVELEX Code 047, within thirty days after completion of testing.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. Unless otherwise specified, preparation for

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delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and the date of this specification.
(b) Selection of applicable levels of preservation, packaging, packing and marking required.

6.2 Preproduction.

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC
(Project 5805-N128)

for termination of wire and ...used personnel at the naval sea
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